product (maker, model, type)
PC (model, speed, ram, hd, price)
laptop (model, speed, ram, hd, screen, price)
printer (model, color, type, price)

1. Find manufactures who sell PC's with the maximum price. (Using RA without MAX)

$R1(model, price1) \leftarrow \pi_{model,\ price}\ (PC)$
$R2(model, price2) \leftarrow \pi_{model,\ price}\ (PC)$
$R3(price) \leftarrow \pi_{price1}\ (R1 \bowtie_{price1 < price2} R2)$
$R4 \leftarrow \pi_{price}\ PC - R3$
$R5 \leftarrow \pi_{maker,\ price}\ (product \bowtie PC)$
$Result \leftarrow \pi_{maker}\ (R4 \bowtie R5)$

2. Find the average price of PC's and laptops made by manufacturer "D". (Using RA)

$R1 \leftarrow \pi_{maker,\ model,\ price}\ (product \bowtie PC)$
$R2 \leftarrow \pi_{maker,\ model,\ price}\ (product \bowtie laptop)$
$R3 \leftarrow R1 \cup R2$
$Result \leftarrow \mathcal{F}_{AVG\ (price)}\ (\sigma_{maker='D'}\ R3)$

3. Find pairs of PC models that have both the same speed and ram. (Using TRC)

$\{\ p1.model, p2.model \mid p1, p2 \in PC \land p1.speed = p2.speed \land p1.ram = p2.ram\ \}$

4. Find the maker of the color printer with a price of at least 120. (Using DRC)

$\{a1 \mid \exists a2,a3\ (<a1, a2, a3> \in product \land \exists b1,b2,b4\ (<b1, b2, b3, b4> \in Printer \land (a2 = b1) \land (b2 = \text{"true"}) \land (b4 > 120)))\}$

5. Find hard-disk sizes that occur in two or more PC's. (Using SQL without COUNT)

```
SELECT DISTINCT p1.hd
FROM PC p1 JOIN PC p2
     ON (p1.model <> p2.model AND p1.hd = p2.hd)
```

6. Find the manufactures of laptops with at least two different speeds. (Using SQL without COUNT)

```
SELECT DISTINCT p1.make
FROM (product p1 JOIN laptop l1 ON (p1.model = l1.model))
      JOIN
     (product p2 JOIN laptop l2 ON (p2.model = l2.model))
      ON (p1.maker = p2.maker AND l1.speed <> l2.speed)
```